

Course: DATA MINING CSE 632 (Fall 2025)

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1. Project 1 title: Website Exploration & Data Mining in Crime Analysis

2. Abstract

In this project, I explored crime incident data from Los Angeles starting in 2020, and I downloaded the data from data.gov. The dataset included information such as the date, time, type of crime, victim's age, and where the crime happened. Before analyzing, I cleaned the data by selecting the most important features, handling missing values, formatting dates and times, and creating new features like day of the week, age groups, and dayparts. I also grouped crime descriptions into categories like violent crimes and white-collar crimes to make the analysis easier.

After preprocessing, I tested three hypotheses about crime trends. The first hypothesis was that white-collar crimes go down in summer while violent crimes increase. The second was that violent crimes are more concentrated in the evening and night, especially among young adults aged 18-29, and that this trend has increased over time. The third was that single-family homes are more likely to be victims of violent crime than multi-family homes. I used bar charts, line graphs, donut charts, and heat maps to visualize the data. These visualizations helped me to see whether the patterns in the data supported or disproved my ideas.

3.1 Problem description, explanation, and data mining in crime data analysis

3.1.1 Crime statistics gathering

When a crime is reported, an LAPD officer files an incident report. That report is entered into the department's Records Management System (RMS). On March 7, 2024, LAPD switched to a new RMS aligned with the FBI's NIBRS standard, which changes how details are coded and sometimes affects update timing.

3.1.2 The official website

The Data.gov catalog page "Crime Data from 2020 to Present" points to the City of Los Angeles open-data portal. It lists CSV, JSON, RDF, and XML downloads plus a landing page link. The Data.gov record shows metadata updated: Aug 23, 2025 and data last modified: Aug 20, 2025.

- **Notes I got from the description:**

- Data comes from original police reports so, some errors are possible.
- Addresses are masked to the nearest hundred block for privacy.
- Missing location can be recorded as (0°, 0°).

These points are stated on the Data.gov dataset page.

- **Landing page or schema:**

The City portal's interactive view shows typical fields like DR_NO, Date Rptd, DATE OCC, TIME OCC, AREA, etc. It provides API access. This is the place to preview the table and export data. I will use the CSV from the official links.

3.1.3 How data mining is used in crime analysis

In my view,

- **Place patterns:** For place patterns, I would put all incidents on a city map. If many points stay close together, that area is a hotspot. I would also use simple clustering to group nearby incidents, so I can say “these places see similar crimes”.
- **Time patterns:** If needed, I would check when crimes happen by hour of day, day of week, and month or season. Then I would compare weekdays vs. weekends and see if the trend is going up or down over the years.
- **Place+time forecasting:** I would try to predict near-future counts. First, I would use simple methods like moving averages or ARIMA. Then I can use grid-based methods that consider both the map and time together, if needed.
- **Feature (what/who/how) patterns:** I would try to look for links between crime type, location, time, premises, weapon, and victim info. I would use basic crosstabs and pivot tables to see which combinations happen more often. For example, there are certain theft types near parking lots in evening.

3.1.4 Three paper summaries

1. Deep learning for real-time crime forecasting:

This paper puts Los Angeles crime counts on a fixed grid [1]. They regularized the data in space and time. Then used a residual CNN (ST-ResNet) to predict short-term crime intensity. Tests on half a year of LA data showed high accuracy. This indicates strong hit rates for top hotspots. This is classic predictive data mining. They took raw incident records, clean and aggregate them into a city grid, then train a supervised model to predict future counts. Then they tuned the model and evaluate with accuracy or hit rate.

2. A systematic review on spatial crime forecasting:

This review collects many studies on spatial crime prediction [2]. It explains which data mining methods people use for crime and compares methods such as statistical, machine learning, hotspot analysis, deep learning methods, and so on. Then it shows that results can change a lot depending on the spatial unit and how the evaluation is done. The authors emphasize how predictive policing can help allocate resources more efficiently and reduce crime rates.

3. Enhancing short-term crime prediction with human mobility flows:

This study adds mobility flow features to historical crime features and builds new features [3]. Then they test several neural models across multiple U.S. cities to forecast next-day risk. This is feature engineering and data fusion. The models that include mobility features improve F1 scores by about 2-7% for next-day predictions. This is a clear example of adding external signals to improve a data mining model.

3.2 Data gathering - how I got the data

3.2.1 Where I started:

I went to the Data.gov page named “Crime Data from 2020 to Present (Los Angeles)”. From there I followed the link to the City of Los Angeles open-data page, the landing page. I chose the CSV option because the assignment asks for CSV.

3.2.2 Download steps:

- i. I opened the landing page
- ii. Clicked Export
- iii. Saved the CSV in my project folder.

3.2.3 Notes I wrote down:

Small warnings from the description like addresses are rounded to the nearest hundred block; some rows may have (0,0) coordinates; reporting system changed in March 2024. Then I used pandas to read a sample (nrows=10000) to make sure dates, and areas look normal.

3.3 Data Exploration

3.3.1 Missing data

I assume “Missing” means NaN or empty. For location I also treat (0,0) as missing.

Feature	Missing count	Missing (%)
LAT, LON (0,0)	2240	0.22
Crm Cd 4	1004927	99.99
Crm Cd 3	1002677	99.77
Crm Cd 2	935831	93.12
Cross Street	850755	84.65
Weapon Used Cd	677744	67.44
Weapon Desc	677744	67.44
Vict Age	138	0.01
Mocodes	151619	15.09
Vict Sex	144644	14.39
Vict Descent	144656	14.39
Premis Desc	588	0.06
Crm Cd 1	11	0.00
Premis Cd	16	0.00
Status	1	0.00

Table 1: Missing data count

3.3.1.1 How I will handle missing data

- LAT, LON (0,0): I will keep all rows and add a flag `has_location` (1 if coordinates are valid, 0 otherwise) so I can report how many records have no coordinates. If later I create any spatial feature, I will use AREA or Rpt Dist No, which are complete. Or, for that step only, I will drop rows without valid coordinates.
- Crm Cd 4: I will ignore this field. I will use the primary crime code only.
- Crm Cd 3: Same as above. I will ignore it.
- Crm Cd 2: Same as ‘Crm Cd 3’ I will focus on the primary code.
- Cross Street: I will not use this column. I will rely on Area or Reporting District or coordinates.
- Weapon Used Cd: I will keep an “unknown or not recorded” group. I will not guess or fill values.
- Weapon Desc: Same as above.
- Vict Age: I will set it to null and later if needed, I will keep an “age unknown” bucket.
- Mocodes: I will use it when present. Otherwise, I will treat it as not recorded.
- Vict Sex: I will keep a “sex = unknown” category. I will not guess.
- Vict Descent: Same as above.
- Premis Desc: If Premis Desc exists, I will map the Premis Cd from the existing other rows; else “unknown”.
- Crm Cd 1: If this is missing, I will use Crm Cd which is complete.
- Premis Cd: If Premis Desc exists, I will map the Premis Cd from the existing other rows; else “unknown”.
- Status: I will use Status Desc if present to map Status; else “unknown”.

3.1.1.2 Not missing, but recorded as unknown

Feature	Unknown count	Unknown %	How I will handle it
Vict Descent	106685	10.62	I will keep “unknown” as a separate bar or line in charts.
Vict Sex	97773	9.73	Same “unknown” stays as its own category.
Status Desc	7	0.00	I will keep it as “unknown”.
LOCATION	16	0.00	Will keep as “unknown”.
Cross Street	18	0.00	Will keep as “unknown”.

Table2: Data recorded as unknown

3.3.2 Noisy features check

We should know what normal looks like before calling something noise. I again looked at the field definitions on the landing page to know what each column means and the common warnings. I ran a small check on pandas to count weird values and duplicates. I applied some simple domain rules given on the landing page for specific columns (time format, age range, victim sex, victim descent). I started with those columns because they have clear rules. For other columns, noise is harder to define without rules.

Rules given on Data.gov page:

1. TIME OCC: In 24 hour military time.
2. Vict Age: Two characters numeric. That means age should be in 00 to 99 years.
3. Vict Sex: F - Female M - Male X - Unknown
4. Vict Descent: Allowed descent W, H, B, A, C, D, F, G, I, J, K, L, O, P, S, U, V, Z, X

Out of these given ranges for the specific features, I counted any other values as noisy. I didn't get any noisy values for these fields.

Features	Count
TIME OCC invalid	0
Vict Age out of range	0
Vict Sex invalid code	0
Vict Descent invalid code	0

Table 3: Noisy value

3.3.2.1 Duplicate Value Check

Duplicate value checking helps identify redundant or repeated entries within a dataset. I also checked if there are any exact full-row duplicates or duplicates by incident id (DR_NO) alone.

Exact full-row duplicates: if there are two rows with identical values for all columns, they should be considered duplicates.

Duplicates by DR_NO: if there are multiple rows with the same DR_NO, but they differ in other details. DR_NO is presumably a unique identifier for incidents.

Field name	Count
Exact full-row duplicates	0
Duplicate DR_NO rows	0

Table 4: Duplicate Value

This indicates there were no exact full-row duplicates, and there were not even duplicate DR_NO in the dataset. Meaning every DR_NO appeared only once in the dataset.

3.3.3 Features that need normalization and discretization

These features can be normalized:

Feature	Why I will normalize it	Normalization I will use	Where I'll use it
Monthly incident counts (from DATE OCC)	Different months have different days, so raw counts are not fair to compare.	Rate per day: $\text{monthly rate} = \text{monthly count} / \text{days in month}$	I can use it in rend lines across months or years.
Area-level counts (by AREA or Rpt Dist No)	Some areas simply have more incidents. I want a fair comparison by areas in the same month.	Rate by area per month = $\text{total incident count in an area} / \text{total incident count in the whole city in same month (as \%)}$.	I will use it if I compare areas over time to see concentration or shifts.
Crime-type counts (by primary Crm Cd / Crm Cd Desc)	Some crime types are more common. If I want to compare rise or fall fairly across months.	Type per month = $\text{crime type count} / \text{total incident count in the whole city in same month (as \%)}$.	When I need to analysis trend by crime type.

Table 5: Features that need

These features can be discretized:

Feature	Why I will discretize it	Buckets I will use
TIME OCC	Parts of the day are easier to compare than 24 separate hours.	Night (0-5), morning (6-11), afternoon (12-17), evening (18-21), late (22-23), plus unknown if time invalid
Vict Age	Age groups make clean charts; I don't impute age.	0-12, 13-17, 18-29, 30-44, 45-64, 65-99, plus unknown
DATE OCC	Shows monthly/seasonal patterns clearly.	Month (jan-dec), quarter (Q1-Q4), or season(winter, spring, summer, autumn).

Table 6: Features that need

3.3.4 Features that need to be created from unstructured data

I will turn a few text columns into small, clean groups so charts are easy to read. Unknown stays “unknown.”

Source column	Features I will make	Rules I will use
Crm Cd Desc	crime family = violent or property or vehicle or fraud or other	Will look for key words. Example: “robbery or assault” - violent; “theft or burglary” - property; “auto or gta” - vehicle; “fraud or identity theft” - fraud; else - other.
Weapon Desc	Weapon group = firearm or knife or blunt or other or unknown	Will find words like gun or pistol or rifle - firearm; knife or blade - knife; bat or club - blunt; empty - unknown.
Premis Desc	Premis group = home or street or parking or business or other or unknown	Will clean the text and map synonyms to common groups. “house or apt” - home; “street or sidewalk” - street; “parking” - parking; “store or restaurant or gas” - business; empty - unknown.

Table 7: Features from unstructured data

3.3.5 Other preprocessing steps

Below are some extra steps that can be done so the data is clean and easy to use.

Steps	Why I need it	What I can do
Rename columns	Mixed cases or spaces are annoying	I will make a clean copy with snake_case names for example date_occ, crm_cd_desc. I will keep the original file untouched.
Fix data types	Some numbers can be read as text	Will check and fix data type errors.
Trim and standardize text	Extra spaces or different cases create fake categories.	I will strip spaces and use uppercase for necessary columns like crm_cd_desc, premis_desc, weapon_desc, status_desc, location
Unify “unknown” tokens	The dataset uses unknown or unk or x	I will not convert “unknown or unk or x” into nan. I keep them as a visible category called unknown.

Table 8: Extra preprocessing steps

3.4 Hypothesis

3.4.1 Hypothesis 1

Hypothesis: White-collar crime for example identity theft, fraud, forgery is going down in summer, while violent crime robbery, assault, rape, homicide, shooting is going up in summer in Los Angeles from 2020 to present.

Why this matters: If violent crime is rising, police and the public need more safety actions; if white-collar crime is rising, we need more prevention for fraud and identity theft. Knowing the direction helps decide where to put time, budget, and awareness.

3.4.2 Hypothesis 2

Hypothesis: Violent crime is more concentrated in the evening and night and especially among young adults (18-29), and this concentration has increased over time.

Why this is important: If more violent incidents happen later in the day and among young adults, patrol schedules and community programs should focus on those hours and age groups. This is a clear temporal pattern the police and the public can act on.

3.4.2 Hypothesis 3

Hypothesis: Single-family homes are more likely to be victims of violent crime than multi-family homes.

Why This Is Important: It's important to know which types of homes are more often targeted for violent crimes because it helps law enforcement and community groups decide where to focus their efforts. For example, if single-family homes are more often victims, the police might need to focus more on neighborhoods with these types of houses. If multi-family homes are targeted more, different strategies might be needed, like adding more security or community programs.

3.5 Preprocessing

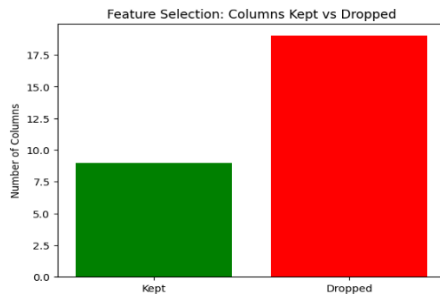
Software tools: For my preprocessing work, I used Python as the programming language and the pandas library for handling the dataset. NumPy helped with numeric operations like defining age bins. They all helped me to reduce the dataset, handle missing values, standardize names, format dates and times, and create meaningful groups like age ranges and dayparts.

3.5.1 Feature selection (keeping only the useful columns)

- **What and how it works:** Feature selection means we keep only the columns that matter for our goals and drop the rest. This makes the data smaller, faster to work with, and easier to understand.
- **How I applied it and parameters:** I wrote a small lookup function to match columns. I kept these columns: case number (DR_NO), dates and times (DATE_OCC, DATE_RPTD,

TIME_OCC), crime code and description (CRM_CD, CRM_CD_DESC), weapon info (WEAPON_DESC), victim age (VICT_AGE), and premise description (PREMIS_DESC). These were chosen because they directly relate to my hypotheses about time trends, age groups, and (optionally) location types.

- **Result**



Category	Count
Kept columns	9
Dropped columns	19
Total columns	28

Figure 1: Columns I kept vs. dropped

Table 9: Total columns vs. I kept vs. dropped

What it says: I reduced the dataset to the exact features needed for my hypotheses.

- **How it helps my hypotheses:** H1 (violent vs white-collar over time): I kept the crime codes/descriptions and dates so we can track trends. H2 (time + ages 18-29): I kept time and age. H3 (home type risk): I kept premise description (to analyze housing types later if needed).

3.5.2 Standardization of column names

- **What and how it works:** Standardizing column names means making them consistent for example: replacing spaces with underscores and using one case. This avoids mistakes and makes code easier to read.
- **How I applied it and parameters:** I converted names to uppercase and replaced spaces with **underscores**. There were no special parameters, just a single rule applied to every column.
- **Result (summarized future figures):**

Column name before	Column name after
DR NO	DR_NO
DATE OCC	DATE_OCC
Date Rptd	DATE_RPTD
TIME OCC	TIME_OCC
Crm Cd	CRM_CD
Crm Cd Desc	CRM_CD_DESC
Weapon Desc	WEAPON_DESC
Vict Age	VICT_AGE
Premis Desc	PREMIS_DESC

Table 10: Column names before and after standardization

What it says: My dataset now has a clean, predictable schema that's easy to use.

- **How it helps my hypotheses:** Cleaner names reduce bugs and confusion later when grouping by year, daypart, age groups, etc. So fewer mistakes when calculating trend lines and comparisons.

3.5.3 Missing Value Handling

- **What and how it works:** Missing value handling means deciding what to do when data is blank. For numbers, I leave NaN so stats or plots can handle them. For text fields, using "UNKNOWN" makes the missingness obvious and countable.
- **How I applied it and parameters:** Numeric columns like VICT_AGE were kept NaN. Text columns like CRM_CD_DESC, PREMIS_DESC fill with "UNKNOWN". This matches our later binning and summaries while staying honest about what we don't know.
- **Result**

Column name	Missing before	Missing after
DR_NO	0	0
DATE_OCC	0	0
DATE_RPTD	0	0
TIME_OCC	0	0
CRM_CD	0	0
CRM_CD_DESC	0	0
WEAPON_DESC	677744	0
VICT_AGE	0	0
PREMIS_DESC	588	0

Table 11: Missing counts per column before vs. after

What it says: We can now measure or filter missing data clearly

- **How it helps my hypotheses:** H2 (age and time patterns): leaving age as NaN prevents fake ages from skewing bins. can still report how many are unknown. H1 or H3: clear "UNKNOWN" labels keep analyses honest. No pretending I know the crime type or premise when we don't.

3.5.4 Date & time formatting (DATE_OCC + TIME_OCC)

- **What and how it works:** Dates and times often come as strings. Formatting converts them into appropriate datetime and time formats so we can safely extract year, month, weekday, and hour without errors.
- **How I applied it and parameters:** I parsed DATE_OCC and DATE_RPTD using the format mm/dd/yyyy hh:mm:ss, with errors="coerce" so bad rows become NaT. Standardized TIME_OCC to a 4-digit string, so hour extraction is reliable.

- **Result**

Category	Before formatting	After formatting
DATE_OCC	07-11-2020	11-07-2020
DATE_RPTD	11-04-2021	04-11-2021
TIME_OCC	845	0845

Table 12: Few date time samples before vs. after formatting

What it says: Dates and times are now standardized and ready for my time-based analysis.

- **How it helps my hypotheses:** H1: I can now group by year and month to compare trends from 2020 to present. H2: reliable time of day bins depend on correct hour parsing.

3.5.5 Data Binning (DAYPART for time, AGE_GROUP for age)

- **What and how it works:** Binning means converting a continuous value like hour or age into labeled buckets like “EVENING” or “18-29”. This makes trends easier to see and compare.
- **How I applied it and parameters**
 - **Daypart (from hour):**
 - morning = 05-11
 - afternoon = 12-16
 - evening = 17-20
 - night = the rest (21-04)
 - **AGE_GROUP (from VICT_AGE):** under_18 (≤ 17), young_adult (18-29), middle_aged_adults (30-44), older_adults (45-64), seniors (65+). I picked these because H2 specifically focuses on 18-29, so that group gets its own bin.
 - **UNKNOWN handling:** If time or age is missing or invalid, I marked the bin as "UNKNOWN" so they're visible in summaries but don't mess valid bins.

- **Result**

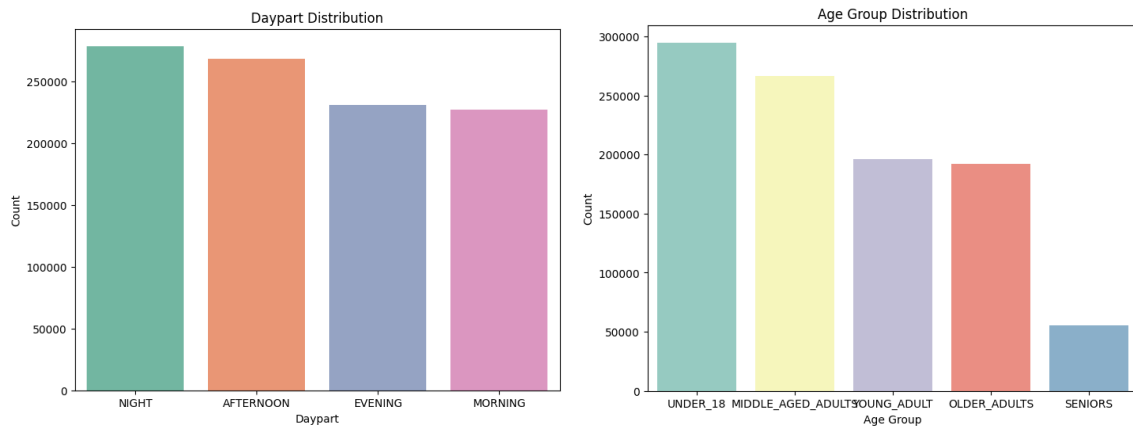


Figure 2: Data count after data binning

- **What it say:** I can now clearly see how incidents spread across times of day and across age groups.
- **How it helps my hypotheses:** H2: Now I can directly test “Are violent crimes more concentrated in evening and night?” and “Is the 18-29 group showing increasing concentration over time?”

3.5.6 Feature making (year, month, weekday from DATE_OCC)

- **What and how it works:** Feature making means creating new, easier to use columns from existing ones so patterns are clear. From the raw DATE_OCC, I made features: year, month, weekday. This turn a long timestamp into simple labels that are perfect for grouping and plotting.
- **How I applied it and parameters:** After converting DATE_OCC into a real datetime, I extracted: year, month, weekday. No special parameters were needed just .dt.year, .dt.month, and .dt.day.
- **Result**

DATE_OCC	YEAR	MONTH	WEEKDAY
11-07-2020	2020	11	SATURDAY
10-18-2020	18	10	SUNDAY

Table 13: New three feature columns from DATE_OCC

- **How this helps the hypotheses:** H1: Year or month let me track counts or rates from 2020 to present and compare trends according to month and year. H2: Pairing year or month with my daypart or age bins lets me see if those concentrations are increasing over time.

3.5.6 Data grouping (crime description grouping - violent vs white_collar vs other)

- **What and how it works:** I scanned CRM_CD_DESC text for words like “ROBBERY or ASSAULT or SHOOTING” to mark VIOLENT, and words like “FRAUD or FORGERY or IDENTITY THEFT” to mark WHITE_COLLAR. Everything else becomes OTHER.
- **How I applied it and parameters:** I made two lists of keywords and assigned the label based on whether the description contained those words. The parameters are simply the keyword lists; they can be tuned if needed.
- **Result:** I will show the figure in next section while proving first hypothesis.
- **How it helps my hypotheses:** H1: Helped me to quickly compare trends for VIOLENT vs WHITE_COLLAR from 2020 to now.

3.5 Visualizing patterns

3.5.1 Hypotheses 1

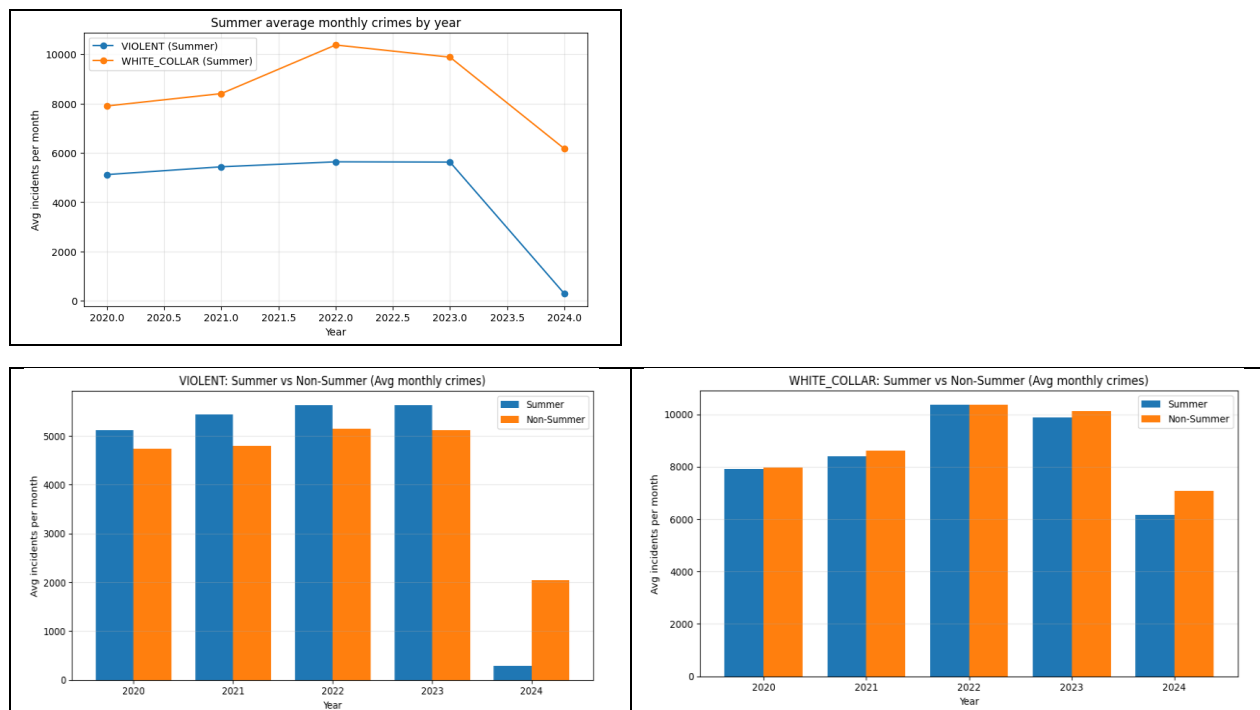


Figure 3: a (top)- summer average monthly crimes by year; b and c (bottom left and right) - Summer vs non-summer by year

Visualization a: I made it to see how summer crime changes over time for violent vs white-collar. I used your year, month, and crime_class, kept only june–aug, counted incidents, divided by 3 months to get the average per month, and drew one line per class. Here, x-axis = year, y-axis = average incidents per month (summer only), blue line = violent (summer); orange line = white-collar (summer); dots show the value for each year. Violent (summer) rises from 2020 to 2023,

then dips in 2024. White-collar (summer) rises to 2022, is similar or slightly lower in 2023, then drops in 2024. The “violent goes up in summer over time” part looks true up to 2023. The 2024 dip is likely because summer is not recorded after that. The “white-collar goes down in summer over time” part is not true for 2020-2023. It went up to 2022, not down. The hypothesis is correct here partly because the violent-crime trend supports it (through 2023), but the white-collar trend does not.

Visualization b and c: I made it to check the seasonal comparison inside each year, if summer is higher (violent) and lower (white-collar) than the rest of the year. For every year*class, I computed avg per month for summer (3 months) and non-summer (9 months), then plotted side-by-side bars. Here, x-axis = year, two bars per year: summer vs non-summer, height = average incidents per month. Violent: in 2020-2023, the summer bar is higher than non-summer each year. 2024 looks odd for the absent data. Seasonal part is supported here as summer is higher than non-summer. White-collar: summer is about the same or slightly lower than non-summer in most full years (2021-2023). Seasonal part is weakly supported here often a bit lower in summer within a year, but not much.

Overall, the hypothesis is partly supported here. This is true for violent crime, not supported for white-collar overtime. So, H1 is not true.

3.5.2 Hypotheses 2

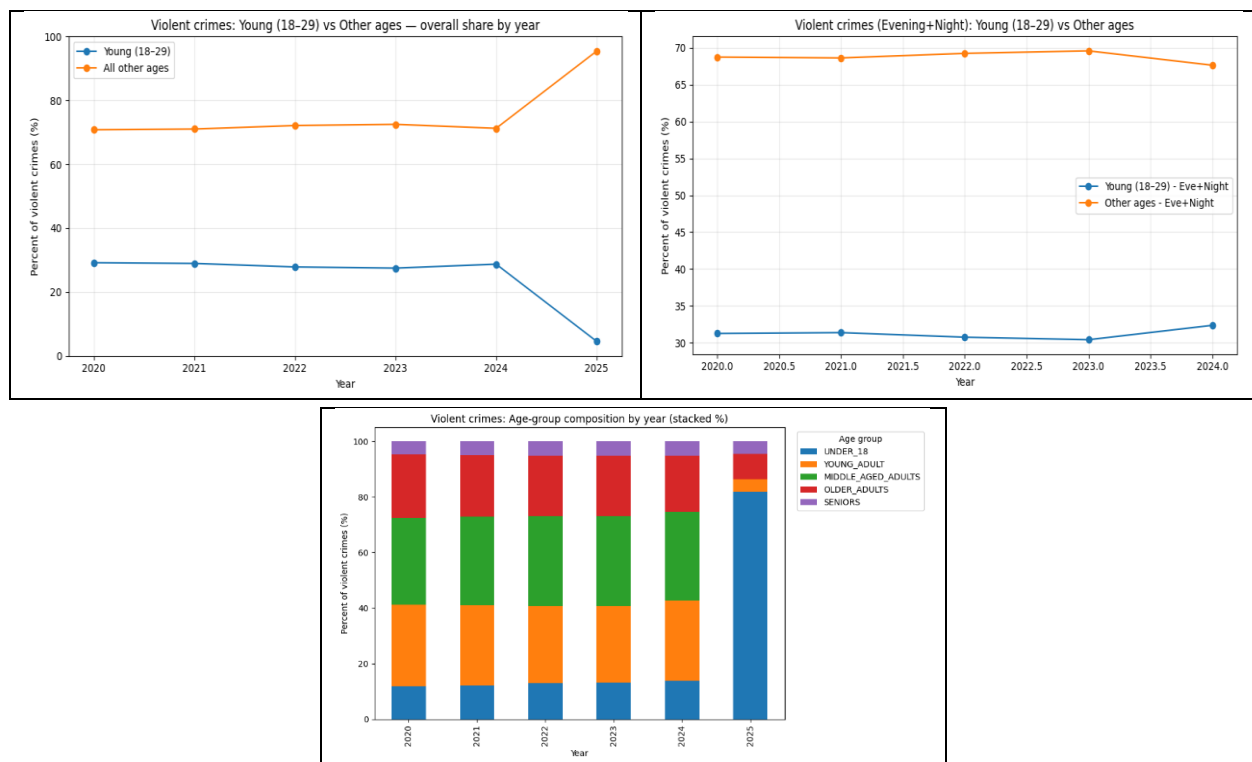


Figure 4: a and b (top left and right) - overall share and evening + night share for young vs other ages respectively; c (bottom) - stacked bar for all age groups by year

Visualization a: I wanted to see whether young adults make up a bigger portion of violent crime compared to other age groups. I created it by grouping violent crime records by year and calculating the percent share of young adults versus everyone else. The x-axis shows years, the y-axis shows percent of crimes, the blue line is for young adults, and the orange line is for all other ages. From 2020 to 2024, young adults stay around 27 to 29% while other ages stay about 70%. In 2025, the young line drops sharply for incomplete data. It shows that young adults are a consistent minority of victims, and their share is not clearly rising. Since the young adult share is mostly flat, this evidence does not prove the hypothesis.

Visualization b: Here I wanted to check whether violent crime among young adults happens more in the evening and night, and if that share grows. I made this plot by filtering crimes in the evening or night, then calculating the share of young adults versus others per year. The x-axis is year, y-axis is percent, the blue line is young adults in evening and night crimes, and the orange line is other ages. Graph patterns show young adults are steady at about 31 to 33%, and other ages stay near 68 to 69%. There is no major increase across time. Young adults do show a concentration in the evening and night, but the share is stable, not rising. According to this graph, the hypothesis is not fully true.

Visualization c: My intention was to compare young adults with all other age groups at once, year by year. I created the graph by counting crimes per age group per year, converting to percentages, and stacking the bars. Here, each bar is a year, and each colored segment is an age group's share of violent crimes. Patterns show that young adults are stable in size from 2020 to 2024. Other groups, like middle-aged adults and older adults, are also steady. 2025 looks unusual because under-18 suddenly dominates due to incomplete data. The distribution does not shift strongly toward young adults over time. The hypothesis is not supported here. The share of young adults is not increasing.

So, hypothesis 2 is only partly correct. The time of day part is supported, but the age group claim is not.

3.5.3 Hypotheses 3

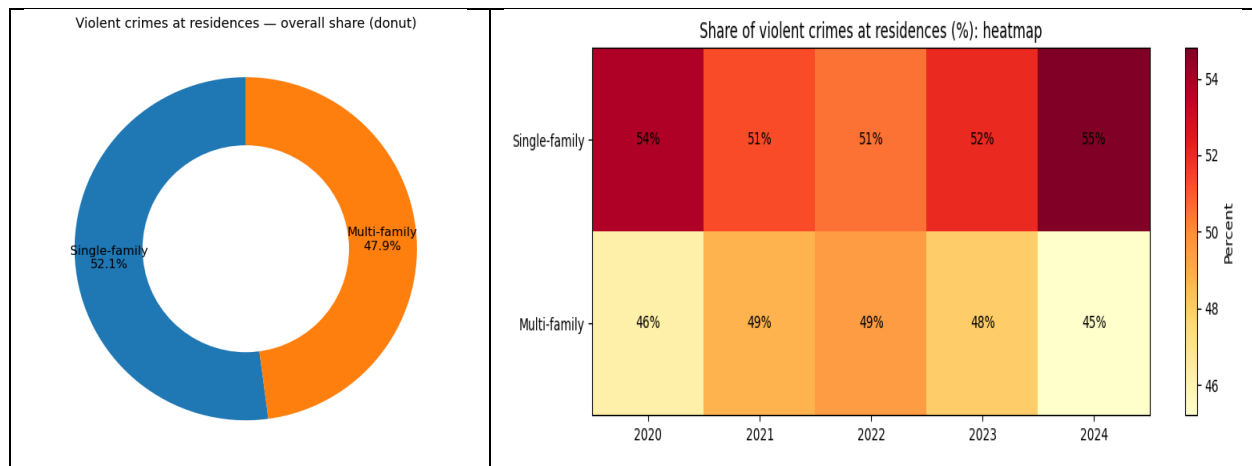


Figure 5: a (left) - overall share; b (right) - home type percentage by year

Visualization a: I created the donut chart to picture the overall picture of violent crimes at residential locations. I grouped all violent incidents into single-family versus multi-family homes and plotted their percentages as slices of the donut. Each slice represents a home type, and the labels show the percent of total residential violent crimes. The chart shows that single-family homes make up about 52%, while multi-family homes account for about 48%. This tells me that violent crimes happen slightly more often at single-family homes overall, which supports my hypothesis.

Visualization b: The heatmap helps me check if this pattern is true every year, not just in total. In this chart, rows are home types, columns are years, and each cell shows the percent share for that year. Darker colors represent higher percentages. From 2020 to 2024, single-family homes are consistently above multi-family homes, staying in the 51-55% range compared to 45-49% for multi-family. This consistency across years gives me confidence that my hypothesis is correct: single-family homes are more likely to be victims of violent crime.

Across the visuals, violent residential crimes occur slightly more often at single-family homes than at multi-family homes. This holds overall, by year, and as a stable trend. Therefore, Hypothesis 3 is supported by the data.

3.5.4 Tools

For this project I used two main software tools, Python and its visualization libraries. For visualizations, I used Matplotlib and Seaborn. Matplotlib allowed me to make bar charts, line graphs, and donut charts by customizing axes, colors, and labels. Seaborn gave me tools for making more polished statistical plots, such as count plots and heatmaps, which highlight patterns in the data with colors.

3.6 Conclusion

My findings partly reflect what we often hear in the news and also match some common real-life experiences. The news often reports that violent crimes, such as assaults or shootings, tend to rise during summer months and at night, which my analysis also confirmed. It also makes sense from experience that young adults are more active during evenings and nights, which could explain why they appear more in the crime data. The result that single-family homes face slightly more violent crimes than multi-family homes is interesting. It shows how living environments may influence risk in different ways.

From these findings, I learned how data can be used to test ideas and not just rely on assumptions. By cleaning the data and visualizing it, I was able to see real patterns that either supported or disproved my hypotheses. This taught me that while news and personal experience can give us clues, data analysis provides stronger evidence for understanding crime trends. It can help communities make better safety decisions.

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